

MLFF common online-monitor specification

Status: current normative monitoring policy

Architecture: revision 105

1. Scope and semantic type boundary

This specification defines deterministic campaign-common **monitoring evidence**. It does not define target-training size, replay-training membership, checkpoint score thresholds, or held-out cross-validation roles.

The current policy families are type-distinct:

OnlineTargetMonitorPolicy

ReplayMonitorPolicy

TargetSizeStudyPolicy

Their records SHALL remain distinct even when two cardinalities happen to share an integer value.

Online monitors never supply gradients.

2. Common target monitor

OnlineTargetMonitorPolicy owns the common target-monitor subset used by authorized development/model-selection procedures, including target-size screening and current checkpoint/stopping control when those consumers explicitly bind this monitor.

The current default requested target-monitor cardinality is exactly:

256 configurations

The current deterministic monitor seed is:

161803

The current default strategy is:

balanced_condition_run_time_systematic

Changing the requested cardinality, seed, strategy, or other policy field changes the policy identity.

Parent role

The target-monitor parent is the DATA5 common outer-monitor/development-monitor domain only. Gradient-training, held-out CV evaluation, calibration, purge-only, excluded, and locked-test roles are not eligible parents.

Deterministic selection

The current target-monitor constructor:

1. groups eligible monitor units by declared condition and source/run identity;
2. allocates the requested budget as evenly as possible across eligible strata under the current deterministic apportionment rule;
3. orders configurations by source trajectory/time within each stratum;
4. uses the policy seed for deterministic random-start systematic sampling;
5. emits exact ordered configuration identities, source indices, stratum available/selected counts, requested/realized size, policy identity, and label-domain identity.

Every fold, seed, training mode, and final-development job using the same compatible campaign monitor identity receives the same target-monitor membership/content identity.

When fewer than 256 eligible configurations exist, all eligible configurations are used and the short-parent fallback is explicit in the record. The realized monitor size remains a monitor property and does not become a target-training size.

3. Independent true-label replay monitor

ReplayMonitorPolicy owns replay-monitor construction. The current default requested replay-monitor cardinality is exactly:

512 configurations

The default replay-monitor strategy is:

chemistry_size_systematic

The true-label replay monitor parent must satisfy the current true-label replay contract. Foundation pseudo labels may be used by a separately identified replay-training path when allowed, but they do not define an absolute true-label replay-validation metric.

Replay-monitor selection uses deterministic ordering across chemistry/composition, atom-count/size grouping, and source order followed by the current systematic selection rule. The materialized replay-monitor artifact is immediately re-inspected; ordered geometry identity, label identity, and true-label mode must match its selection record.

When fewer than 512 eligible true-label replay configurations exist, all are used and the fallback is explicit.

Replay training and replay monitoring are different evidence roles and may not silently alias one another.

4. Record contracts

OnlineTargetMonitorRecord

Binds at least:

- target-monitor role;
- parent-domain digest;
- OnlineTargetMonitorPolicy digest;
- requested and realized sizes;
- exact ordered selected identities/source indices;
- per-stratum available/selected counts;
- strategy and seed;
- label-domain identity;
- explicit fallback reason where applicable.

ReplayMonitorRecord

Binds at least:

- replay-monitor role;
- replay source/label lineage;
- ReplayMonitorPolicy digest;
- requested and realized sizes;
- exact ordered selected identities/source indices;
- strategy/seed fields owned by the replay policy;
- true-label mode where required;
- materialized artifact digest where applicable;
- explicit fallback reason where applicable.

TrainingProtocolIdentity binds the exact monitor record/policy identities it uses. It does not infer them from filenames.

5. Leakage and independence invariants

- Monitoring configurations never contribute gradients.
- The common target monitor is development/model-selection evidence, not held-out CV evidence.
- Held-out CV evaluation and locked tests cannot be promoted into online-monitor roles.
- Target-training geometries may not overlap true-label replay-monitor evidence where the current replay contract forbids such overlap.
- Membership is selected once per compatible campaign identity and reused; it is not redrawn per epoch, fold, seed, or target-size candidate.
- The monitor seed is policy identity, not an informal hint.
- A monitor cardinality is never interpreted as N_{selected} .

6. Relationship to target-size study

TargetSizeStudyPolicy may consume the common target and replay monitors as authorized development/model-selection evidence.

The target-size study does not own their construction and cannot change their membership between size candidates. Every size candidate/seed sees the same monitor identities so comparison is paired with respect to evaluation evidence.

The target-size ladder remains the fixed population owned by `mlff_target_subset_size_study_spec.md`; monitor sizes 256 and 512 are semantically independent.

7. Precision and accumulation boundary

Monitor inference uses the learned model precision/backend declared by TrainingProtocolIdentity. mdstats-owned metric accumulation remains under the current numerical-precision specification and is not weakened by model dtype.

Changing model precision/backend creates a different training/runtime protocol identity where the current architecture declares it protocol-defining.

8. Persistence and unsupported historical records

Current monitor artifacts are accepted only when their current schema, content, parent lineage, and policy digests validate.

Obsolete monitor/campaign schemas do not gain current meaning through compatibility aliases. A campaign whose monitor records cannot validate under the current generation requires re-preparation.

9. Acceptance requirements

Current monitor qualification covers at least:

1. deterministic target/replay membership regeneration;
2. exact requested defaults and explicit short-parent fallback;
3. common target-monitor identity across compatible fold/final/seed jobs;
4. condition/run/time distribution rather than first-N truncation;
5. true-label replay enforcement where required and exact artifact round-trip;
6. forbidden-role/geometry-overlap checks;
7. protocol identity binding;
8. corruption/staleness rejection;
9. proof that monitor cardinalities are not consumed as target-size authority.